

Now I have successfully cloned a new project called hostel from erb7. The hostel project has the same structure and business logic like erb7. However, the hostel project has its own Postgres database named hostel. I just received the first requirement for the project hostel but it is written in Chinese. Here is the full requirement:

“Requirement (I)

- 為至少三個 Django 資料模型建立不少於 20 筆樣本資料記錄。
- 產生一個獨立的 Python 程式碼來：
 - a) 清理原始資料
 - b) 格式化資料集
 - c) 能將資料匯入和匯出到 Django 資料庫
 - d) 最終結果可以在 Django 管理面板下檢查

”

Please:

1. Translate the Requirement (I) above into English with explanations.
2. List all steps to fulfill Requirement (I).

Thanks for checking, SIU! You're absolutely right—my previous reply cut off after Step 4. Let's go back and give you the **complete, detailed response** to fulfill **Requirement (I)** for your **hostel Django** project.

1. English Translation of Requirement (I) with Explanation

◆ Original Chinese:

Requirement (I)

- 為至少三個 Django 資料模型建立不少於 20 筆樣本資料記錄。
- 產生一個獨立的 Python 程式碼來：
 - a) 清理原始資料
 - b) 格式化資料集
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 - d) 最終結果可以在 Django 管理面板下檢查

◆ English Translation:

Requirement (I)

- Create **at least 20 sample data records** for **at least three Django models**.
- Develop a **standalone Python script** that can:
 - **(a)** Clean the raw data
 - **(b)** Format the dataset
 - **(c)** Import and export data to/from the Django database

- (d) Ensure the final result is visible and verifiable in the **Django admin panel**

✓ 2. Full Steps to Fulfill Requirement (I)

◆ Step 1: Identify or Create 3 Django Models

- Choose or define at least three models in `hostel/models.py`. Example:
- ```
class Room(models.Model):
 name = models.CharField(max_length=100)
 capacity = models.IntegerField()
 floor = models.IntegerField()

class Resident(models.Model):
 full_name = models.CharField(max_length=100)
 check_in_date = models.DateField()
 room = models.ForeignKey(Room, on_delete=models.CASCADE)

class Booking(models.Model):
 resident = models.ForeignKey(Resident, on_delete=models.CASCADE)
 start_date = models.DateField()
 end_date = models.DateField()
 status = models.CharField(max_length=50)
```
- Register these models in `hostel/admin.py`:
- ```
from django.contrib import admin  
from .models import Room, Resident, Booking  
      
admin.site.register(Room)  
admin.site.register(Resident)  
admin.site.register(Booking)
```

◆ Step 2: Create Sample Data (≥ 20 records per model)

- Use Python's `faker` library or manual entries
- Create a script or Django management command to insert sample data

Example using `faker`:

```
from faker import Faker  
from hostel.models import Room, Resident, Booking  
import random  
fake = Faker()  
  
# Create 20 rooms  
for _ in range(20):  
    Room.objects.create(  
        name=fake.word().capitalize(),  
        capacity=random.randint(1, 4),  
        floor=random.randint(1, 10)  
    )
```

Queries to Als_20251017(五)

Repeat for Resident and Booking.

◆ Step 3: Write a Standalone Python Script (data_tool.py)

This script should include:

a) Data Cleaning

- Remove duplicates
- Drop nulls or invalid entries
- Normalize inconsistent formats

Example:

```
Room.objects.filter(name__isnull=True).delete()
```

b) Data Formatting

- Standardize casing, date formats, etc.

Example:

```
for resident in Resident.objects.all():
    resident.full_name = resident.full_name.title()
    resident.save()
```

c) Import/Export to CSV

Export:

```
import csv
from hostel.models import Room

with open('rooms.csv', 'w', newline='') as file:
    writer = csv.writer(file)
    writer.writerow(['Name', 'Capacity', 'Floor'])
    for room in Room.objects.all():
        writer.writerow([room.name, room.capacity, room.floor])
```

Import:

```
with open('rooms.csv', newline='') as file:
    reader = csv.DictReader(file)
    for row in reader:
        Room.objects.create(
            name=row['Name'],
            capacity=int(row['Capacity']),
            floor=int(row['Floor'])
        )
```

◆ Step 4: Verify in Django Admin Panel

- Run your Django server:
- `python manage.py runserver`
- Log in to `http://localhost:8000/admin`
- Check that:
 - All three models appear
 - Sample data is visible
 - Imported/exported data is correctly formatted

✅ Summary of What You Need

Task	Action
Define models	Create at least 3 Django models
Add sample data	≥ 20 records per model
Build script	<code>data_tool.py</code> for cleaning, formatting, import/export
Admin check	Verify results in Django admin panel

If you'd like, I can help you write the full `data_tool.py` script or set up a management command version. Want to automate sample data generation next?

-END-